ISTQB FOUNDATION MASTERCLASS

Boundary Value Analysis Cheat Sheet

Find the defects that hide at the edges of every input range.

Introduction

Boundary Value Analysis (BVA) is a black-box test technique based on the observation that defects cluster at the edges of equivalence partitions. Instead of testing every value, you test the values on and around each boundary. This cheat sheet gives you the rules, a worked example and the exam traps.

Why Boundaries Matter

Programmers make mistakes with relational operators (< versus <=), off-by-one errors and loop conditions. These bugs surface precisely at the boundaries of a range, so testing boundary values gives high defect-detection power for very few test cases.

Two-Value vs Three-Value BVA

Approach	Values tested per boundary	Coverage
Two-value (2-point)	The boundary value and its nearest neighbour on the other side	Standard ISTQB Foundation approach
Three-value (3-point)	The value before, on, and after the boundary	Stronger; recommended when risk is high

Worked Example — an integer field valid for 1 to 100

The valid partition is 1..100. Its boundaries are 1 (lower) and 100 (upper). Below is the full set of two-value and three-value boundary values.

Boundary	Two-value BVA	Three-value BVA
Lower (1)	0, 1	0, 1, 2
Upper (100)	100, 101	99, 100, 101

So for two-value BVA you would test 0, 1, 100 and 101. Values 0 and 101 are invalid; 1 and 100 are valid. This tiny set catches almost every off-by-one defect in the field.

Step-by-Step Method

1. Identify the equivalence partitions for the input (valid and invalid).
2. Find the minimum and maximum boundary of each partition.
3. Select the boundary value and its immediate neighbour(s).
4. Design one test case per boundary value, noting the expected result.
5. Remember: a valid boundary and the adjacent invalid value are both worth testing.

Exam Traps & Common Mistakes

- Confusing the boundary with the whole partition — BVA tests the edges, Equivalence Partitioning tests a representative from the middle.
- Forgetting that boundaries exist on both valid and invalid partitions.
- Using the wrong increment for non-integers (for a currency field to 2 decimals, the neighbour of 100.00 is 100.01, not 101).
- Assuming three-value BVA is the ISTQB default — the syllabus default is two-value.

WATCH OUT

The neighbouring value must respect the field precision. For a field accepting one decimal place valid 0.0 to 9.9, the value just above the upper boundary is 10.0, and the value just below is 9.8 for three-value BVA — not 10 and 9.

Best Practices

- Combine BVA with Equivalence Partitioning — they are complementary.
- Always record the expected result for each boundary case.
- Escalate to three-value BVA for safety-critical or high-risk boundaries.

INSIDE STLC PRO TIP

In interviews and the ISTQB exam, if asked "how many test cases for a 1–100 field using two-value BVA", the answer is four: 0, 1, 100 and 101. Say it with confidence.